



DETECTION OF FAKE JOB POSTINGS USING MACHINE LEARNING

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Literature Review

Kumar et al. (2019): focused on text features based on TF-IDF and Logistic Regression and showed their baseline model with word-level text patterns could signal fraudulent job postings.

Sharma et al. (2020): advocated for utilizing deep learning with LSTMs to capture latent semantic meaning in job descriptions to enhance detection of fraud in job postings as they capture hidden patterns in long data instances.

Rastogi et al. (2021): stressed that class imbalance was a significant problem as fraud is a small percentage (~5%) of postings. They showed that oversampling and weighting models were important steps towards better classification.

Project Objectives

- Detect fraudulent job postings by combining textual features (descriptions, job requirements) with structured information (employment type, company logos) .
- Tackle the dataset imbalance (approx 5% fraud) using techniques including oversampling, undersampling or cost-sensitive models.
- Train and compare machine learning models (Logistic regression, random forest, Xgboost, Svm, LSTM) to discover the most effective fraud detection approach.
- Provide insights for safer recruitment, including common characteristics of fake job postings (short/vague job descriptions, lack company logos, unrealistic offers).

Proposed System Modules

The workflow consists of the following components:

Preprocessing – Involves cleaning the text, encoding categorical variables, and dealing with missing values.

Feature Engineering – Use TF-IDF, embedding, and metadata (logo, job type, description length).

Model Training – Using Logistic Regression, Random Forest, XGBoost, SVM, and LSTM. Evaluation – Tools include Accuracy, Precision, Recall, F1, and ROC-AUC.

Prediction – Classification as Real or Fake.

Modules: Data Collection → Preprocessing → Feature Extraction → Model Building → Evaluation → Prediction

Conclusion

Fake job postings are an increasing menace, as they entice and deceive searching job seekers.

Machine learning models can accurately detect fake job postings while dealing with issues such as class imbalance.

The model alerts the system about true fraudulent job postings and allows Detectable to identify patterns of fraud (e.g. vague job description, no company information).

Future Scope: To commercialization by deploying this work as a web application or browser plugin that detects fraud in real-time on the Internet.



THANK YOU!

